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Game theory and its application to Politics

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Abstract

This paper provides a systematic analysis of the applications of game theory in political science, illustrating how the strategic decision-making framework of game theory can be employed to illuminate and improve political outcomes. This thesis conducts a comprehensive review of the literature, examining the use of game theoretical models in various political domains, including international relations, electoral politics, political negotiations, public policy, and institutional design. This analysis reveals how game theory can be utilized to predict political behavior, design effective policies, and facilitate cooperation in political contexts. Also, identify areas for future research and provide recommendations for policymakers and political analysts seeking to leverage game theory in their work. By providing a comprehensive overview of the intersection of game theory and political science, this paper aims to serve as a valuable resource for scholars and practitioners seeking to harness the insights of game theory to improve political decision-making.

Declaration

I, Nkhoma Ernest, hereby declare that the work contained in this research project is my original work, and that any work done by others or by myself previously has been acknowledged and referenced accordingly.

Author's Signature: ..... Date: ...../...../.....

Supervisor's Signature: ..... Date: ...../...../.....

Dedications

I proudly dedicate this piece of research work to mum and auntie; Ms. Elizabeth Nyangulu and Dr. Stella Chisanga Zulu for the love, inspiration, care and sacrifice they have given me, I'll forever appreciate. To my siblings, Hastings, Gloria, Macsensio, Augustine, and Cynthia Nkhoma I feel so lucky and honored to have you and call you my brothers and sisters no matter how down I feel you people are always there to cheer me up, I love you people. To my best friend Annastazia Soko, thank you so much for being my tower of strength ever since we knew each other you have always been encouraging me to become better, you are irreplaceable.

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1 Introduction

Game theory, the science of strategic decision-making, has long been applied to various fields from economics and biology to psychology and computer science [18]. However its potential to illuminate the complex dynamics of political party interactions remains largely untapped. Political parties, as rational actors, engage in strategic games to achieve their goals, influence policy outcomes, and maximize their political payoffs. Yet,the intricacies of these interactions, shaped by conflicting interests, ideological differences, and institutional constraints , remain poorly understood.
In the realm of democratic systems, the interplay between two dominant political parties is particularly crucial, as it can significantly impact policy outcomes , voter behavior , and the overall political landscape . By applying game theory to the analysis of political party interactions, we can uncover the strategic dynamics that drive their decision-making processes, revealing the optimal strategies, Nash equilibria , and Pareto optimal outcomes that emerge from their competing interests.

Linear programming (LP), on the other hand, a subset of mathematical optimization offers a robust method of optimizing decision-making process . By formulating political party interactions as linear programs, we can quantify political party preferences and utility functions, optimize political party strategies for maximizing payoffs, predict policy outcomes and political negotiations and evaluate the stability and efficiency of political systems.

This research project seeks to bridge the gap between game theory, LP and political science, to provide a comprehensive and qualitative analysis exploring the applications of game-theoretic concepts and LP framework, unlike opinion polls which mostly results into rigging and eventually causing chaos and wars in countries, This thesis also aim to provide a novel perspective on political party behavior, shedding light on the strategic interactions, negotiations, and bargaining processes that shape democratic systems.

Through this inquiry, this study hope to contribute to more deeper understanding of political party dynamics, policy outcomes and democratic process as whole.

1.1 History of Game Theory

Discussions on the mathematics of games began long before the rise of modern mathematical game theory. Cardano's world Liber de ludo aleae(book on game of chance) which was written around 1564 but published posthumously in 1663, sketches some basic ideas on games of chance [5].

Girolamo Cardano

Liber de Ludo Aleae

1563

Figure 1.1: Cardano

The discipline of game theory was pioneered in the early 20th century by the mathematicians Ernst Zermelo (1913) and John Von Neumann (1928). The breakthrough came with john Von Neumann and Oskar Morgenstern's book, theory of games and economic behavior published in 1944 [22]. This followed important work by John Nash and Lloyd Shapely.
Game theory had a major influence on the development of several branches of economics (industrial organization, international trade, labor economics, microeconomics, etc.). Over time the impact of game theory extended to other branches of the social sciences (political science, international relations, philosophy, sociology, etc.) as well as fields outside the social science such as, biology, computer science, logic, etc,.

Game theory became closely associated withe Rand Corporation after the second World. The Rand corporation was a private company spun off from the US Air Force at the end of the war and it was specially warfare. Game theory was obviously relevant to this task and as a result the organization championed its development by hiring as a consultant Von Neumann and other central figures in development of game theory, like John Nash, Duncan Luce and Howard Raiffa. In 1950, game theory developed dramatically, when John Nash developed game theory tools and concepts

for general **non-cooperative theory** and **cooperative bargaining theory**. He introduced what is now called the 'Nash equilibrium' of a strategic game in 1951. Game theory was broadened theoretically and applied to problems of war, political science and philosophy in the 1950's and 1960's. British philosopher R.B. Braithwaite in his book of **The theory of Games as a tool for the Moral philosopher** was applied to philosophy in 1955. In this book, Braithwaite provides how games can be used to arrive at moral and ethical decisions. As a specialized filed game theory was established by great mathematicians and economists, in the last quarter of the 20th century and early 2000s. Game theory acknowledged special attention in 1994 with the awarding of Nobel memorial prize to three game theorists; john Nash, John Harsany and Reihard Selten, for their theoretical work in game theory which was very influential in economics. At the same time, the US Federal Communications Commission was using game theory t held design $7 billion auction of the radio spectrum for personal communication services (naturally the bidders used game theory too).

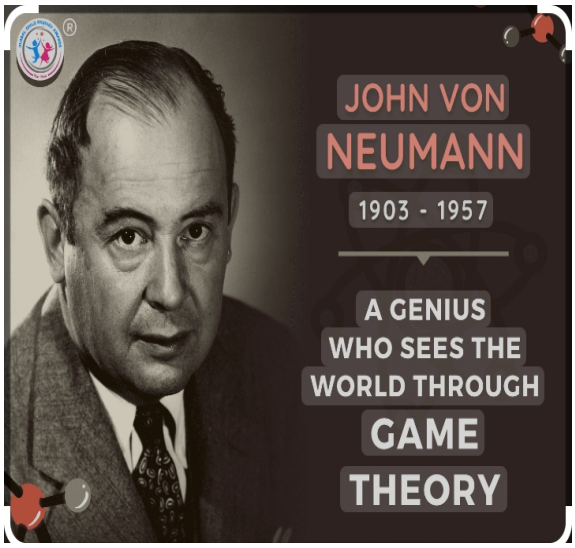

Figure 1.2: John Von Neumann

1.2 Background of study

Game theory's application to politics emerged in the mid-20th century, as scholars sought to analyze political behavior and strategic interactions using mathematical tools. However, it was not until 1960s that game theory began to be applied to political science, with the work of scholars such as Thomas Schelling, William Riker, and Kenneth Arrow. Since then, the field has grown exponentially, with the contributions from scholars across various disciplines, including political science, economics and sociology. Its applications have been diverse, ranging from international relations to domestic politics. In international relations, game theory has been used to study topics such as nuclear deterrence, arms races and international cooperation.

In domestic politics, game theory has been used to the study of electoral competition, political negotiations and institutional design. Game theory's concepts, such as Nash equilibrium, Pareto optimally, and the Prisoner's dilemma have became essential tools in understanding political behavior.
Despite its contributions, the application of game theory to politics is not without limitations. Critics argue that game theory that game theory oversimplifies political reality, neglects contextual factors and relies heavily on rational choice assumptions.

1.3 Statement of the problem

Game theory has the potential to revolutionize international relations, but how can political leaders effectively negotiate and cooperate in a world of conflicting interests and limited trust?.
The challenge lies in finding strategies that balance national interests with global cooperation, while navigating the complexities of diplomatic relationships and geopolitical dynamics.
In the realm of political campaigns, winning is everything but what strategies can campaigns employ to maximize their chances of success? The electoral landscape is fraught with uncertainty, and campaigns must navigate the intricacies of voter behavior, electoral competition, and campaign finance to emerge victorious.
When it comes to public policy, the stakes are high, and the consequences of failure can be severe. How can policymaker design and implement effective policies that take into account the strategic interaction and incentives of various stakeholders? The answer lies in applying game theory to understand the complex dynamics of policy-making and develop optimal strategies for success.
Legislative politics is a game of numbers, but how can political actors achieve their goals in a world of coalition formation and voting behavior? The key lies in understanding the strategic interactions between lawmakers, parties, and interest groups, and developing effective strategies to build coalitions, negotiate, and vote.
Corruption is a pervasive problem in politics, but how can game theory be applied to mitigate its risks and promote transparent and accountable governance? By understanding the incentives and strategic interactions that drive corrupt behavior, policymakers can develop effective strategies to prevent it and promote good governance. Conflict is a natural part of politics, but how can political leaders manage and resolve disputes effectively? The answer lies in understanding the complexities of human psychology and group dynamics, and developing strategies that take into account the strategic interactions incentives of all parties involved.

1.4 Aims

(i) To investigate the applications of Game theory to Zambian politics.

(ii) Investigate game theory's role in optimizing policy-making and governance in Zambia.

1.5 Objectives of the study

1.5.1 Specific objectives

(i) Review the literature of applications of game theory in military, business, economics and politics (in USA, Africa, Ghana, Zambia)

(ii) Review the methodology of game theory

1.5.2 Overall objectives

(i) Evaluate the effectiveness of different political party strategies and policy outcomes using game-theoretic and linear programming(LP) approaches.

(ii) Formulate a linear programming model to allocate resources(e.g., campaign funds, voter turn out) and investigate the impact of different objective functions.

1.6 Significance of study

This study holds significant importance as it pioneers an innovative integration of game theory and linear programming to optimize political party interactions and policy outcomes, thereby enhancing the efficiency and effectiveness of democratic governance. By developing a comprehensive framework for analyzing and predicting political outcomes, this research contributes to the advancement of political science, game theory and LP, while providing valuable insights for political parties, policymakers, and scholars.The study findings have far-reaching implications for improving political decision-making, promoting democratic stability,and informing political strategy, ultimately leading to more effective policy making processes and better government.Furthermore, this piece of writing demonstrates the potential of interdisciplinary approaches to address complex real-world problems, paving the way for further studies to explore the applications of game theory, LP in various fields. By shedding light on the intricate dynamics of political party interaction and policy outcomes, this study empowers political actors and scholars to make more informed decisions, fostering a deeper understanding of democratic system and their optimization.

2 Literature Review

2.1 Applications of game theory

2.1.1 Introduction

Game theory has profound impact and extensive applications across a wide of fields, including economics, political science, business, military computer science, biology, sports, etc. and providing a robust frame work for analyzing and understanding strategic decision-making, predicting behavior, and optimizing outcomes in complex interactive solutions, enabling researchers and practitioners to develop innovative solutions, strategies, and policies that address real-world problems, improve performances and enhance overall well-being, and its applications continue to grow and evolve as new areas of study emerge, further solidifying its position as a fundamental tool for understanding and navigating complex systems and interactions. This chapter briefly explains the application of game theory in business, military and defense, economics and politics

2.1.2 Military and Defense

The application of game theory in defense has sustained and diversified history ranging from the design of real-time military systems(e.g., applied in missile interception) to the support of strategic decisions on large defense investments and acquisitions. Game theory enhances military strategies and decision making processes with a holistic and quantitative analysis of situations. For the military, the potential scenarios amenable to game-theoretic analysis include the rapidly growing applications of autonomous intelligent systems and game theory provides a comprehensive mathematical framework that greatly enhances the capabilities of decision making of the people who use these systems. Game theory often finds application in national security and anti-terrorism related fields. This includes predicting and preparing for terrorist attacks, as well as resource allocation scenarios for the protection of key personnel and landmarks/other potential targets for terrorist activity. While the value of potential targets and the likelihood of attacks are obviously key parameters governing the payoff functions of games in this niche, the subsequent social , economic and political ramifications are equally instrumental in modeling games in this area. Game theory models analyzes cyber threats, developing strategies for defense attacks [8]. it protects military network and systems from cyber attacks, ensuring secure communication exchange [9]. It aids in understanding the motivation and behavior of adversaries, predicting their actions and identifying vulnerabilities. It informs intelligence gathering and analysis, helping military forces to stay one step ahead of potential threat. Game theory also optimizes resource allocation, supply chain management, and transportation planning in military logistics . it ensures efficient and effective support for military operations [12].

2.1.3 Business

Game theory, a mathematical framework for analyzing strategic interactions between players, has revolutionized the way business approach decision-making. By applying this theory of games, companies can optimize their strategies, anticipate competitor's actions, and achieve better outcome, we will explore the various applications of game theory in business, from strategic planning to international trade and diplomacy.

(i) Strategic planning
Game theory helps companies anticipate and prepare for competitor's actions, developing strategic plans and policies . by analyzing the interaction the interaction between players, game theory provides a framework for understanding the strategic implications of different actions and reactions. this enables businesses to develop effective counter-strategies, negotiate with suppliers and customers, and optimize their overall strategy [20].

(ii) Pricing and revenue management
Game theory informs pricing decisions, analyzing competitor's pricing strategies and optimizing revenue . Game theory helps businesses determine the optimal price point, taking into account factors such as market demand, competition, and customer behavior

(iii) Supply chain management
Game theory optimizes supply chains operations, negotiating with suppliers, and managing inventory. It also helps businesses determine inventory levels, negotiations and optimize their overall supply chains.

(iv) Human resource and organizational design
Game theory informs human resource(HR) policies and organizational design, managing employee incentives , also helps businesses determine the optimal HR policies, taking into account factors like productivity and job satisfaction.

(v) International trade and diplomacy
Game theory develops strategies for international trade and diplomacy, negotiating with foreign governments. It determines the optimal trade strategy, taking into account factors such as tariffs, quotas, and market access.

2.1.4 Economics

The most common application of game theory to economics includes one player's strategy affecting the other players' choices and vice versa. Game theory is an essential part of economics because it is one of the few fields that examine rational

human behavior and predict how people will react to given conditions.
Studying game theory in economics also help us understand why some countries are prosperous while others are not, enabling us to learn development strategies and better policies. In economics, game theory applications explain how the decisions made by individuals or firms affect others in the market.

In *behavioral economics*, game theory studies people's incentives and actions to create the best outcome for themselves. It is important to learn game theory because, many times, financial results depend on a mix of altruism and self-interest (according to economists). Understanding what causes someone to behave rationally is key. Most economists have come with three main approaches to modeling behaviors. One is Nash equilibrium which suggests players will always want to choose the most efficient option available. The second is iterated elimination of dominated strategies, which means if they are multiple options, players will eliminate those less advantageous until they only have two options left. The third approach is stochastic dominance, which claims humans always want to choose the outcome that is most likely or has highest probability of success.

The main application of game theory in economics is its use in forecasting market behavior. Forecasting market behavior helps experts create more efficient supply chains. When making decisions whether or not they should enter certain markets or invest money into inventory models, many forecasting strategies are used by firms.

Here are few prominent examples of game theory in economics.

(i) A classic game theory problem is the **Stag Hunt**. Suppose there are two hunters in a clearing, one is stalking a deer, and the other must wait for 30 minutes before they can start hunting. If they both hunt simultaneously, neither will catch anything, but if one hunts while the other waits, whoever catches it gets all the meat for themselves . But, what should each person do, given the other might not cooperate?
There are four possible outcomes:

  (i) They both hunt together
  (ii) One hunter decides to wait
  (iii) Only one person hunts
  (iv) Neither hunts

  What is the best for each individual depends on their expected payoff under each situation. This payoff is calculated as their personal share of the meat multiplied by how likely they think they will get to keep it (given that their counterpart cooperates).
  For instance, if you think your partner has more than a 50% chance of cooperating with you when you decide to hunt (which would give you 10 points), then waiting while your partner hunted would give you 5 points instead of 0 points from not hunting.

(ii) To fully grasp what is game theory in economics, it is first necessary to understand the **Voting paradox**. the voting paradox works as follows;
Voters are asked to rank their preferences for one of two candidates, A and B. They are then allowed to vote as many times as they want, ranking A first and B second. For instance one voter might give their rankings as ABBABBABABAB, indicating that voter prefers candidate a over candidate B. Still, if it were not the case, the rankings would be recorded so that there would be a higher preference for candidate B(BBAAAB). Votes cast by this system should theoretically allow a to be determined mathematically.

(iii) **Prisoner's dilemma:**[23] It can be said that prisoner's dilemma is the highly disputed and inspected game in International Relations discipline. Prisoner's Dilemma tackles the interstate cooperation and common action problems. According to Liberals, in international relations, there are two ways to improve the cooperation: hegemon powers assume to provide public property in order to ease all states in the system and parties come near for cooperation by long-term considering as a result of repetitive interactions. The using of prisoner's dilemma helps us understand the contribution of repetitive interactions to the cooperation. Since, Liberals, in contrast to realists, denote that self-interests will actually prompt actors to the cooperation in the long term. According to the notion named shadow of the future, it can be shown those actors who do not approach to the cooperation or who choose conflict due to their self-interest, tight in short term, to the extent that they approach to the cooperation, they can acquire more profits in long term. According to the story of the game, the two suspects were taken into custody by the police as part of an investigation. Since there is not enough evidence for the arrest in the hands of the police, he put both suspects in separate cells and offers an agreement. According to the agreement, if one of the suspects testifies against the other, and the other remains speechless, the party who prefers to remain free to testify will be sentenced to 10 years in prison [19]. If they both remain silent when they do not testify against each other, both will be sentenced to 1 year in prison, and if both testify against each other, both suspects will be sentenced to 5 years in prison [17]. In this context, both suspects have to choose between witnessing or staying silent. Both suspects are not allowed to learn the decision of the other until the end of the investigation, that is, the two suspects in different rooms are not able to communicate with each other. Accordingly, the player, who is unaware of the opponent's decision, will not remain silent, unable to afford the possibility of 10 years in prison, and will testify against the other party. By witnessing against the other party, he will settle for a shorter prison term of 5 years or be released. Here the player will aim to minimize his loss (maximize his winnings). It will be inevitable that the other side will testify under the same conditions by acting rationally. Thus, the decision taken by the two parties who do not communicate with each other by acting rationally rather than in good faith actually leads to an increase in prison terms, which they may face less. Assuming that both players prefer the short term prison sentence over the long one, and each of them will not gain any benefit by shortening the other's penalty, the prisoner's dilemma is

included in zero-sum games. As in game theory, the sole purpose of the players in this game is to maximize their winnings independently from the other player. Accordingly, strategic level game matrix can be shown as follows:

	Prisoner B	
Prisoner A	confess	Stay silent
Confess	A(5,5)	B(0,10)
Stay silent	C(10,0)	D(1,1)

Table 2.1: Prisoners dilemma Game Matrix

The remarkable part of this game is that the option to confess for both players is the dominant strategy. Thus, whatever the other player chooses, confessing will bring more preferable punishments. There are two interesting points about the nature of international relations in this game. The first is that the strategies that the two actors are rational and pursue their own interests are not optimal. Due to the lack of communication between them and the distrust of each other, prisoners choose the way of confessing with fear of the other's confession. If they cooperate, they would receive only one year penalty (cell D), the equilibrium in this game is that they receive they will receive ten years penalty (cell A). Many international situations are also like this; rational choices can lead actors to conflict and pareto-ineffective consequences. The use of this game to understand international relations is due to this dilemma. This dilemma examines why actors in international relations tend to conflict and develop less useful policies for themselves, and sometimes why they choose cooperation. Second interesting point is that prisoner's dilemma demonstrates how structure affects the behaviors of actors. If the rules of game (structure) were different, actors could find more chance and possibility for cooperation. Prisoner's dilemma is used in the literature to model problems that require many interstate common actions, such as armament races and environmental pollution prevention [14]

2.1.5 Politics

In the realm of politics, game theory provides critical insights into how individuals and groups make decisions in competitive environments. by modeling situations where the outcome for each participant depends on the action of others game theory provides the understanding of political behavior, including voting strategies.
Political actors often face choices that involves uncertainty and interdependence, making game theory a valuable tool for understanding phenomena like conflict and pursuit of power.

Many scholars and analyst in different countries across the globe have applied game theory to predict an election outcome. Some notable countries which applied game theory to predict an election are discussed hereunder;

(i) **USA:** Game theory has been extensively applied in American politics to analyze and predict strategic interactions between political actors, including

politicians, parties, interest groups, and voters. Its principles have been used to explain various phenomena, such as election strategies, legislative bargaining, voting behavior, party competition, and international relations. For instance, the Median Voter Theorem, a game theory model, demonstrates how candidates adjust their platforms to appeal to swing voters while maintaining support from their party's base, as seen in Mitt Romney's 2012 presidential campaign. Similarly, game theory concepts like Nash equilibrium help explain how lawmakers negotiate and compromise on policy, exemplified by the 2013 budget negotiations between Democrats and Republicans.

Game theory also informs understanding of voter decision-making, including the **paradox of voting**, where rational voters may abstain due to perceived low impact. However, models like the **calculus of voting** show how voters weigh factors like civic duty and policy preferences. In party competition, game theory examines how parties position themselves relative to each other, as demonstrated by the 1990s' triangulation strategy employed by Democrats like Bill Clinton. Furthermore, game theory guides analysis of U.S. foreign policy interactions, such as nuclear deterrence and trade agreements, exemplified by the 2015 Iran nuclear deal.

Scholars have also applied game theory to study cabinet and Supreme Court appointments, analyzing strategic interactions between the President, Senate, and nominees, as seen in President Obama's successful nomination of Sonia Sotomayor. Additionally, game theory informs discussions on public policy debates, evaluating strategic interactions among stakeholders in areas like healthcare reform, climate change, and tax policy. Notable scholars, such as Anthony Downs and William Riker, have pioneered game theory applications in American politics, influencing policymakers, political scientists, and economists. Their work enhances understanding of strategic interactions shaping American politics, from electoral campaigns to legislative negotiations.

Recent research has expanded game theory's scope, incorporating behavioral and psychological factors to better explain political decision-making. This includes studies on voter biases, partisan polarization, and the impact of social media on political discourse [7]. Moreover, advances in computational methods have enabled researchers to simulate complex political scenarios, predicting outcomes and testing hypotheses. As game theory continues to evolve, its applications in American politics will provide valuable insights into the strategic dynamics underlying political institutions, processes, and outcomes. By integrating game theory with other disciplines, scholars can develop more comprehensive theories of political behavior, ultimately informing policy decisions and improving governance. Game theory is playing a crucial role in predicting outcomes, analyzing candidate strategies, and identifying key swing states in the 2024 US presidential election.
Election forecasting models, such as The 538 Forecast Model, The Economist's Election Tracker, and The ABC News/Washington Post Poll, utilize game theory to simulate different scenarios and predict the likelihood of each candidate winning [6]. These models take into account various factors, including polling

data, demographic information, and historical trends. Predictive maps, like 270 to win and Poly market, provide interactive tools that allow users to create their own forecasts. These maps consider polling data, demographic information, and other factors to predict electoral votes. Game theory concepts, such as Nash Equilibrium and Signaling Theory, are being applied to analyze candidate strategies and voter decision-making. Researchers are using game theory to study how candidates allocate resources, choose campaign themes, and respond to opponents' moves.

Some notable predictions based on game theory and forecasting models include Harris leading in Pennsylvania, a tight race in North Carolina, and electoral vote projections favoring Harris. However, it's essential to recognize that these predictions are subject to change and should be taken as probabilistic forecasts rather than certainties.

Game theory also informs candidate strategy analysis, including resource allocation and debate strategies. By analyzing voter behavior, game theory identifies key demographics and issue-based voting patterns. Additionally, game theory models predict voter participation rates [21]. Despite its applications, game theory faces limitations and challenges in election forecasting. Uncertainty and the dynamic nature of elections can affect predictive accuracy. Future directions include integrating machine learning to enhance predictive accuracy and incorporating real-time data analysis.

(ii) **Africa:** African politics presents a complex and dynamic landscape, characterized by diverse cultures, economies, and political systems. Hence African politics faces numerous challenges such as; Ethnic and religious conflicts ,Electoral disputes and democratic transitions, Natural resource management and economic development, Regional integration and cooperation, External influences and global governance etc. The following are some of the African countries to have applied game theory to their political arena.

(i) **South Africa :** South Africa's transition from apartheid to democracy (1990-1994) is a notable example of game theory in action. Nelson Mandela's African National Congress (ANC) and F.W. de Klerk's National Party engaged in strategic negotiations, employing game theoretical principles such as: Signaling: Mandela's release from prison and de Klerk's speech acknowledging ANC as a legitimate partner, Bargaining: Power-sharing agreements and constitutional reforms, Coalition formation: ANC's alliance with smaller parties [16].

(ii) **Ghana :** Ghana, a democratic country in West Africa, has successfully applied game theory in various political contexts. In 2012, Ghana's Supreme Court utilized game theoretical principles to resolve electoral disputes, leveraging concepts like the prisoner's dilemma to understand incentives for cooperation among political parties. The court's ruling sent signals to parties about electoral integrity, while negotiations between parties and the Electoral Commission demonstrated bargaining principles[15]. Ghana's political parties have also employed game theory in coalition formation. In 2004, the New Patriotic Party (NPP) and

National Democratic Congress (NDC) formed coalitions using coalitional game theory, analyzing party alliances and identifying stable coalition structures. This strategic interaction led to peaceful transitions and enhanced democratic governance.Ghana's constitutional review process in 2011 also employed game theoretical principles. Public consultations signaled commitment to democratic reforms, while negotiations among stakeholders shaped constitutional provisions. Key players, including the Electoral Commission, Supreme Court, and civil society organizations, have contributed to Ghana's effective application of game theory. The outcomes have been significant, enhancing electoral integrity, promoting peaceful transitions, improving natural resource management, and strengthening democratic institutions. Future applications of game theory can inform policy decisions on economic development, enhance regional cooperation and integration, and improve public sector governance. [1]

(iii) **Zambia:** Since gaining independence in 1964, Zambia's political landscape has exhibited strategic decision-making, mirroring game theoretical principles. Pre-independence negotiations between Kenneth Kaunda's United National Independence Party (UNIP) and British colonial authorities demonstrated tactical interactions. Post-independence, UNIP's dominance led to strategic decisions on maintaining power, while economic management involved tactical choices on nationalization and planning.

Zambia's transition to multiparty democracy in 1991 introduced electoral competitions, where parties like UNIP, Movement for Multi-Party Democracy (MMD), Patriotic Front (PF), and United Party for National Development (UPND) engaged in strategic campaigning. Coalition formations, pre-election alliances, and post-election negotiations reflected bargaining principles. Researchers have identified implicit game theory applications, including strategic voting, signaling, coalition formation, and Nash equilibrium.

Notable events, such as the 1991 transition, 2006 constitutional reform, 2009 presidential by-election, and 2021 general election, have showcased Zambia's evolving political strategy. Scholars like Dr. Mwanga Chibiliti and Dr. Kelvin Mtezo have analyzed Zambia's politics through a game-theoretic lens, providing valuable insights. While game theory hasn't been explicitly applied, its principles have underpinned Zambia's political decisions, informing electoral strategies, coalition building, and economic management.

3 Methodology

3.1 Introduction

Any business involves competition. Effective decision making plays a vital role in such situation mainly because decisions will have direct impact on the revenue earning potentials of business organization.
This chapter explains the terminologies used in game theory, i.e. Player, Strategy, Payoff matrix, Maximin principle, Saddle point, Minimax principle, Value of the game, Two person zero-sum game. Decision making situations may be classified into the following three types: Deterministic situation, Probabilistic situation and uncertainty situation.

Deterministic Situation. Under this situation, the problem data representing the situation are constants. They do not vary with respect to time or any other basis.

Probabilistic Situation. Under this situation, the problem data representing the situation do not remain constant. They vary due to chance, and it is possible to present the data in the form of some probability distribution. Later, these probability distributions become components of decision making model of the situation.

Uncertainty situation. When the problem data are subjected to variation and it is not possible to represent them in the form of any probability distribution, this situation is called uncertain situation. Game theory is an example of the uncertainty situation.

Game theory is the branch of mathematics concerned with strategies for dealing with competitive situations where the outcome of one participant's choice of action depends "critically" on the other participant's choice of actions [22].

3.2 Terminologies of Game Theory

The different terminologies that are associated with game theory are now defined.

(i) Player
There are two players in a game. The players may be two companies (for example, company A and company B) competing for tenders, two countries planning for trade gains in a third country, two persons biding in a game, etc.

(ii) Strategy
It is a course of action taken by a player, for example, giving computer furniture free of cost, giving 20% additional hardware, giving special price, etc., while selling computer hardware. The strategy can be further be classified into pure strategy and mixed strategy. Let m be the number of strategies for player A. n be the number of strategies of player B, p_i be the probability of selection of the alternative i of player A, for i= 1, 2, 3,....., m., q_j be the probability of selection of the alternative j of player B, for j= 1, 2, 3,.....,n.

The sum of the probabilities of selection of various alternatives of each player is equal to 1 as shown below.

$$\sum_{i=1}^{m} p_i = 1$$

$$\sum_{j=1}^{n} q_j = 1$$

(i) Pure Strategy
If a player selects a particular strategy with a probability of 1, then that strategy is known as **pure strategy**. This means that the player is selecting that particular strategy alone ignoring his remaining strategies.
If player A follows a pure strategy, then only one of the p_i values will be equal to 1 and the remaining p_i values will be equal to 0. A sample set of probabilities of selection of the alternatives for player A is shown below:

$p_1 = 0$, $p_2 = 1$, $p_3 = 0$.

The sum of these probabilities is equal to 1. That is;

$p_1 + p_2 + p_3 = 0 + 1 + 0 = 1$

(ii) Mixed strategy
If a player follows more than one strategy, then the player is said to follow a **mixed strategy**. Buy the probability of selection of te individual strategies will be less than one and their sum will be equal to one. If player B follows a mixed strategy, then a sample set of probabilities 0f selection of mixed strategy is shown below:

$q_1 = 0.65$, $q_2 = 0$, $q_3 = 0.35$

It is clear that the sum of the probabilities is equal to 1. That is:.

$q_1 + q_2 + q_3 = 0.65 + 0 + 0.35 = 1$

(iii) Payoff matrix $[a_{ij}]$
Each combination of the alternatives of players A and B with outcome, a_{ij}. If a_{ij} is positive, then it represents a gain to player A, and a loss to player B. If it is negative, it represents a loss to player A and a gain to player B. A sample payoff matrix of player is as shown below.

$$
\begin{array}{c|c|cccccc}
 & & \multicolumn{6}{c}{\text{player}B} \\
 & & 1 & 2 & \cdots & j & \cdots & n \\
\hline
\text{player A} & 1 & a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\
 & 2 & a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\
 & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 & i & a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{in} \\
 & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 & m & a_{m1} & a_{m2} & \cdots & amj & \cdots & a_{mn}
\end{array}
\tag{3.1}
$$

(iii) Maximin Principle
This principle maximizes the minimum guaranteed games of player A. The minimum gains with respect with to different alternatives of A, irrespective of B's alternatives are obtained first. The maximum of these minimum gains is known as the **maximin value** and the corresponding alternative is called as **maximin strategy**.

(iv) Minimax principle
This principle minimizes the maximum losses. The maximum losses with respect to different alternatives of player B, irrespective of player A's alternatives, are obtained first. The minimum of these maximum losses is known as the **minimax value** and the corresponding alternative is called as **minimax strategy**.

(v) Saddle point
In a game, if the maximin value is equal to the minimax value, then the game ia said have a **saddle point**. The intersecting cell corresponding to these values is known as the **saddle point**. If the game has a saddle point, then each player has a **pure strategy**.

(vi) Value of the game
If the game has a saddle point, then the value te value of the cell at the saddle point ia called the **value of the game**; otherwise, the value of the game is computed based on expected value calculations which will be explained later.

(vii) Two-person zero-sum game
In a game with two players, if the gain of one player is equal to the loss of another player, then that game is called **two-person zero-sum game**.

3.3 Game with pure strategy

As we know, if a game has a saddle point, then the game is said to have a pure strategy for each of the players. Determination of such pure strategy of each of the players is illustrated in the following example.

Example 3.3.1.

Two computer hardware manufacturing companies, company A and company B, competing for supplying computers to a government department. Each company has listed has its strategies for selling computers.

Strategies for company A

(i) Giving special price

(ii) Giving 20% worth of additional hardware

(iii) Supplying computer furniture free of charge

Strategies for company B

(i) Giving special price

(ii) Giving 30% worth of additional hardware

(iii) Giving free training to the users of the organization which is buying hardware

The estimated gains(+)/losses(-) of company A for various possible combinations of the alternatives of both companies are summarized in the table. In the table the rows represent the alternatives of company A and the columns represent the alternatives of company B. Each cell value of the table represents the the estimated gains(+)/losses(-) of company A in lakhs of rupees for the corresponding alternatives of company A and B. A positive cell entry denotes the gain to company A(loss to company B) and a negative cell entry denotes the gain loss to company A(gain to company B). The table is called as the payoff matrix with respect to company A. Determine the optimal strategy/strategies for company A and company B.

	Company B		
Company A	1	2	3
1	20	15	22
2	35	45	40
3	18	20	25

Table 3.1: Payoff matrix for company A

Solution: If company A selects strategy 1 and company B selects strategy 2, the corresponding pay-off to company A ia Rs. 15 lakhs. This means that company B would be losing Rs. 15 lakhs. In table 2, all the cell entries are positive . so, whatever may be the selected strategy of company B, company A will always a gain.

But the magnitude of the gain varies for different combinations of rows and columns. Though company B has no chance of winning the game, it can try to minimize the gain of company A, so that in the long run, company B will have a competitive advantage. If there are some negative cell entries in table 2, then there is a chance of winning the game for company B, provided company A selects any strategy any of the strategies corresponding to those negative cell values.
As per payoff matrix shown in table 2, company A is called as **maximin player** and company b is called as **minimax player**, which means company A is maximizing its minimum guaranteed gain and company b is minimizing its maximum loss. Table 2 is reproduced wit necessary calculations, where,
Maximin value= Minimax value= 35

Hence, the game has a saddle point at the cell corresponding to row 2 and column 1. The value of the game is Rs. 35 lakhs.
The optimal probabilities of selection of strategies of company A and company B are as given below.

A$(p_1, p_2, p_3) = A(0, 1, 0)$ and B$(q_1, q_2, q_3) = B(1, 0, 0)$

	Company B			
Company A	1	2	3	
1	20	15	22	15
2	$\boxed{35}$	45	40	35(max)
3	18	20	25	18
	35(min)	45	40	

Table 3.2: Payoff matrix of player A with minimum and maximum values

Company A should always select its second strategy with a probability of 1 and it should select all other strategies with zero probability. Company b should always select its first strategy with a probability of 1 and should select all other strategies with zero probability.
If any one of the two companies deviates from its optimal strategic combination(s), the other player will have a gain. This means that company A should always select its second alternative (giving 20% worth of additional hardware). If it deviates from its alternative, company B will gain in terms of its reduced loss, since all the entries of table 2 are positive values (loss for company B will be less than K35)
As per optimal solutions, company b should always select its alternative 1 (giving a special price). If it deviates from thus strategy, company A will have an additional gain over and above the value of the game . Under this situation, the gain for company A will bw more than K30.
[1]

[1]If one of the companies follows the optimal solution as per the theory of game and the other company does not use any such guidelines but follows a random strategy, definitely the first company will be in an advantage because the second company deviates from optimal strategy, as determined using the theory of game.

3.4 Games with mixed strategy

If the game has no saddle point, then the game is said to have mixed strategies. To demonstrate the determination of mixed strategies; consider the 2×2 payoff matrix with respect to player A which has no saddle point.

	B	
A	1	2
1	a	b
2	c	d

Table 3.3: A game with mixed strategy

Algorithm to determine mixed strategy

step 1: Find the absolute value of a-b(i.e.$|a-b|$) and write it against row 2.

step 2: Find the absolute value of c-d(i.e. $|c-d|$) and write it against row 1.

step 3: Find the absolute value of a-c(i.e. $|a-c|$) and write it against column 2.

step 4: Find the absolute value of b-d(i.e. $|b-d|$) and write it against column 1.

The results of the above steps are summarized in the table below. The absolute values are called as *oddments*

	B		
A	1	2	oddments
1	a	b	$\|c-d\|$
2	c	d	$\|a-b\|$
oddments	$\|b-d\|$	$\|a-c\|$	

Table 3.4: A game with mixed strategy

step 5: Compute the probabilities of selection of the alternative of player A(p_1 and p_2) and that of player B(q_1 and q_2).

$$p_1 = \frac{|c-d|}{|a-b|+|c-d|}$$

$$p_2 = \frac{|a-b|}{|a-b|+|c-d|}$$

$$q_1 = \frac{|b-d|}{|a-c|+|b-d|}$$

$$q_2 = \frac{|a-c|}{|a-c|+|b-d|}$$

The value of the game can be computed using any of the following formulae:

$$V = \frac{a|c-d|+c|a-b|}{|a-b|+|c-d|}$$

$$= \frac{b|c-d|+d|a-b|}{|a-b|+|c-d|}$$

$$= \frac{a|b-d|+b|a-c|}{|a-c|+|b-d|}$$

$$= \frac{c|b-d|+d|a-c|}{|a-c|+|b-d|}$$

Example 3.4.1.

Consider payoff matrix with respect to player A and solve it optimally.

	B		
A	1	2	
1	6	9	
2	8	4	

Solution: The maximin and minimax values of the given problem are shown in the table below.

	B		
A	1	2	
1	6	9	6(max)
2	8	4	4
	8(min)	9	

In this problem, the maximin value (6) is not equal to the minimax value (8). Hence, the game has no saddle point. Under this situation, the formulae given in this section are to be used to find the mixed strategies of the players and also te value of the game. The computations of oddments of the game are summarized below.

	B		
A	1	2	oddments
1	6	9	4
2	8	4	3
oddments	5	2	

Table 3.5: payoff matrix with oddments

Let p_1 and p_2 be the probabilities of selection of alternative 1 and alternative 2, respectively of player A. Also q_1 and q_2 be the probabilities of selection of alternative 1 and alternative 2 of payer B, respectively. Then, we have ;

$$p_1 = \frac{|c-d|}{|a-b|+|c-d|} = \frac{4}{3+4} = \frac{4}{7}$$

$$p_2 = \frac{|a-b|}{|a-b|+|c-d|} = \frac{3}{3+4} = \frac{3}{7}$$

$$q_1 = \frac{|b-d|}{|a-c|+|b-d|} = \frac{5}{2+5} = \frac{5}{7}$$

$$q_2 = \frac{|a-c|}{|a-c|+|b-d|} = \frac{2}{2+5} = \frac{2}{7}$$

Where the value of the game is,

$$V = \frac{a|c-d|+c|a-b|}{|a-b|+|c-d|} = \frac{(6)(4)+(8)(3)}{4+3} = \frac{48}{7}$$

Hence the strategies of player A is: A($\frac{4}{7}$, $\frac{3}{7}$) and of player B is ;B($\frac{5}{7}$, $\frac{2}{7}$). The value of the game is $\frac{48}{7}$.

3.5 Dominance Property

In some games, it is possible to reduce te size of the payoff matrix by eliminating redundant rows (or columns). If a game has such redundant rows(or columns), those rows or columns are dominated by some other rows(or columns), respectively. Such property is known as **dominance property**.

(i) Dominance property for rows
(a) In the payoff matrix of player A, if all the entries in a row *(X)* are greater than or equal to the corresponding entries of another row *(Y)*, then row Y is dominated by row X. Under such situation , row Y of the payoff can be deleted.
(b) In the payoff matrix of player A, if each of the sum of the entries of any two rows (sum of the entries of row X and row Y) is greater than or equal to corresponding entry of the third row *(Z)*, then row Z is dominated by row X and row Y. Under such situation, row Z of the payoff matrix can be deleted.

(ii) Dominance property for columns
(a) In the payoff matrix of player A, if all the entries in a column *(X)* are less than or equal to the corresponding entries of another column *(Y)*, then column Y is dominated by column X. Under such situation, column Y of the payoff can be deleted.
(b) In the payoff matrix of player A, if each of the sum of the entries of any two columns (sum of the entries of column X and column Y) is greater than or equal to corresponding entry of the third column *(Z)*, then column Z is dominated by row X and row Y. Under such situation, column Z of the payoff matrix can be deleted.

Example 3.5.1.

Players A and B plays a game in which each player has three coins($K20$, $K25$, $K50$). Each of them selects a coin without the knowledge of the other person. If the sum of the values of the coins is an even number, A wins B's coin. If the sum is an odd number B wins A's coin.

(i) Develop a payoff matrix with respect to player A.

(ii) Find the optimal strategies for the players.

Solution: The payoff matrix with respect to player A is shown in the table below. The maximin and minimax values are also indicated in the table. The maximin value (-20) is not equal to the minimax value (20). Hence the game has no saddle point. As a result, the game as mixed strategies.

	Player B			
Player A	I K20	II K25	III K50	
I K20	20	-20	50	-20
II K25	-25	25	-25	25(max)
III K50	20	-50	50	-50
	20(min)	25	50	

Check for dominance property: Row 3 is dominated by row 1 and row 3 is to be deleted. The resultant matrix after deleting roe 3 is shown below.

Player A	Player B		
	I K20	II K25	III K50
I K20	20	-20	50
II K25	-25	25	-25

Table 3.6: payoff matrix after deleting row III

In the table 3.5, the column III is dominated by column I, and hence, column III is to be deleted. The resultant matrix after deleting column III is shown below. The oddments of the rows and columns are presented in the same table.

Player A	Player B		
	I K20	II K25	oddments
I K20	20	-20	50
II K25	-25	25	40
oddments	45	45	

Table 3.7: Payoff matrix with oddments

Let p_i be the probability of selection of the alternative i by player A, where i= 1, 2, and q_j be the probability of selection of alternative j by player B, where j= 1, 2. Then we have;

$$p_1 = \frac{50}{50+49} = \frac{5}{9}$$

$$p_2 = \frac{40}{50+40} = \frac{4}{9}$$

$$q_1 = \frac{45}{45+45} = \frac{1}{2}$$

$$q_2 = \frac{45}{45+45} = \frac{1}{2}$$

Where, te value of the game is;

$$V = \frac{50(20) - 25(40)}{50+40} = 0$$

Hence the strategies of player A is: A($\frac{5}{9}$, $\frac{4}{9}$, 0) and of player B is ;B($\frac{1}{2}$, $\frac{1}{2}$, 0). The value of the game is 0.

Example 3.5.2.

Consider a 4×4 game played by players A and B, and solve it optimally .

	Player B			
Player A	1	2	3	4
1	6	2	4	8
2	2	-1	1	12
3	2	3	3	9
4	5	2	6	10

Table 3.8:

Solution: In the table , the computations for finding the maximin and minimax values of the given game are shown. The maximin value (2) is not equal to the minimax value (3). Hence, the game has has no saddle point.

	Player B				
Player A	1	2	3	4	
1	6	2	4	8	2(max)
2	2	-1	1	12	-1
3	2	3	3	9	2(max)
4	5	2	6	10	2 (max)
	6	3(min)	6	12	

Table 3.9: Payoff matrix for minimax and maximin values

Check for dominance property: In table, for each row, the sum of the values in column 1 and column 2 is lesser than or equal to the corresponding value in column 4. Hence column 1 and column 2 dominate column 4. So, column 4 is to be deleted and the corresponding result is sown below.

	Player B		
Player A	1	2	3
1	6	2	4
2	2	-1	1
3	2	3	3
4	5	2	6

Table 3.10: Payoff matrix after deleting column 4

Here, for each column, the sum of the values in row 1 and row 3 is greater tan or equal to the corresponding value in row 2. Hence, row 2 is to be deleted and the corresponding result is shown below.

	Player B		
Player A	1	2	3
1	6	2	4
3	2	3	3
4	5	2	6

Table 3.11: Payoff matrix after deleting row 2

Here, the values in column 2 are less than or equal to the corresponding values in column 3. Hence column 2 dominates column 3. So, column 3 is deleted, as shown below.

	Player B	
Player A	1	2
1	6	2
3	2	3
4	5	2

Table 3.12: Payoff matrix after deleting column 3

Now, the values in row 1 are greater than or equal to to the corresponding values in row 4. Hence, row 1 dominates row 4. So row 4 ia deleted.

	Player B		
Player A	1	2	oddments
1	6	2	1
3	2	3	4
oddments	1	4	

Table 3.13: Payoff matrix after deleting row 4

Here, the oddments are shown. Based on these values, the probabilities of selection of the alternatives that are present are computed.
Let p_i be the probability of selection of the alternative i by player A, where i= 1, 2, and q_j be the probability of selection of alternative j by player B, where j= 1, 2. Then

$$p_1 = \frac{1}{1+4} = \frac{1}{5}$$

$$p_2 = \frac{4}{1+4} = \frac{4}{5}$$

$$q_1 = \frac{1}{1+4} = \frac{1}{5}$$

$$q_2 = \frac{4}{1+4} = \frac{4}{5}$$

$$V = \frac{6(1) - 2(4)}{1 + 4} = 2.8$$

The optimal solution to the problem is:
A($\frac{1}{5}$,0, $\frac{4}{5}$, 0) ;B($\frac{1}{5}$, $\frac{4}{5}$, 0, 0), $V = 2.8$.

3.6 Graphical methods for 2×n or m×2 Games

Some games will be of specialized nature. Like 2×n or m×2 games. the payoff matrix of the 2×n game will contain 2 rows and n columns, whereas the payoff matrix of the m×2 game will contain m rows and 2 columns. If there is no saddle point for these games, one can solve them using graphical method.

Algorithm for 2×n game
step 1: Reduce the size of the payoff matrix of player A by applying dominance property, if it exists.

step 2: Let x be the probability of selection of alternative 1 by player A and $1 - x$ be the probability of selection of alternative 2 by player A. Derive the expected gain function of player A with respect to each of the alternatives of player B.

step 3: For each of the gain functions which are derived in step 2, find the value of the gain, when x is equal to 0 as well as 1.

step 4: Plot the gain functions on a graph by assuming a suitable scale. Keep x on X-axis and gain on Y-axis.

step 5: Since the player A is a maximin player, find the higher intersection point in the lower boundary of the graph. Let it be the maximin point.

step 6: If the number of lines passing through the maximin point is only two, form a 2×2 payoff matrix from the original problem by retaining only the column corresponding to those two lines and go to *step 8*; otherwise go to **step 7**.

step 7: Identify any two lines with opposite slopes passing through that point. Then form a 2×2 payoff matrix from the original problem by retaining only the columns corresponding to those 2 two lines which are having the opposite slopes.

step 8: Solve the 2×2 game using oddments and find the strategies for player and B and also the value of game.

Algorithm for m×2 game

step 1: Reduce the size of the payoff matrix of player A by applying dominance property, if it exists.

step 2: Let y be the probability of selection of alternative 1 by player B and $1-y$ be the probability of selection of alternative 2 by player B. Derive the expected gain function of player B with respect to each of the alternatives of player A.

step 3: For each of the gain functions which are derived in step 2, find the value of the gain, when y is equal to 0 as well as 1.

step 4: Plot the gain functions on a graph by assuming a suitable scale. Keep y on X-axis and gain on Y-axis.

step 5: Since B is a minimax player, find the lowest intersection point in the upper boundary of the graph. Let it be the minimax point.

step 6: If the number of lines passing through the maximin point is only two, form a 2×2 payoff matrix from the original problem by retaining only the rows corresponding to those two lines and go to *step 8*; otherwise go to **step 7**.

step 7: Identify any two lines with opposite slopes passing through that point. Then form a 2×2 payoff matrix from the original problem by retaining only the rows corresponding to those 2 two lines which are having the opposite slopes.

step 8: Solve the 2×2 game using oddments and find the strategies for player and B and also the value of game.

Example 3.6.1.

Consider the payoff matrix of player A and solve it optimally using graphical method.

	Player B				
Player A	1	2	3	4	5
1	3	0	6	-1	7
2	-1	5	-2	2	1

Table 3.14:

Solution: The table below shows the payoff matrix with minimax and maximin values. Here, the maximin(-1) is not equal to the minimax (2). Hence the game has no saddle point, as a result the players will have mixed strategies. Since the game has only two rows, it can be solved using graphical method.

	Player B					
Player A	1	2	3	4	5	
1	3	0	6	-1	7	-1(max)
2	-1	5	-2	2	1	-2
	3	5	6	2(min)	7	

Table 3.15: Payoff matrix with minimax and maximin values

Checking the dominance property: In the payoff matrix shown in the table above, column 5 is dominated by column 1. Similarly, column 2 is dominated by column 4. Hence, delete column 2 and column 5. The resultant payoff matrix is shown below.

	Player B		
Player A	1	3	4
1	3	6	-1
2	-1	-2	2

Table 3.16: Payoff matrix after deleting column 2 and column 5

Let, x be the probability of selection of alternative 1 by player A and $1-x$ be the probability of selection of alternative 2 by player A. Therefore the expected payoff to player A with respect to different alternatives of player B is summarized below .

B's alternative	A's payoff function
1	$3x-(1-x)=4x-1$
3	$6x-2(1-x)=8x-2$
4	$-x+2(1-x)+-3x+2$

Table 3.17: Expected functions of player A

The computations of the expected payoff of player A to each of the alternatives of player B, when $x=0$ as well as when $x=1$ are summarized below.

B' alternative	A's expected payoff function	A's expected gain $x=0$	$x=1$
1	$4x-1$	-1	3
3	$8x-2$	-2	6
4	$-3x+2$	2	-1

Table 3.18: Expected gain of player A

Now , the expected gain functions of player A with respect to different alternatives of player B are plotted below.

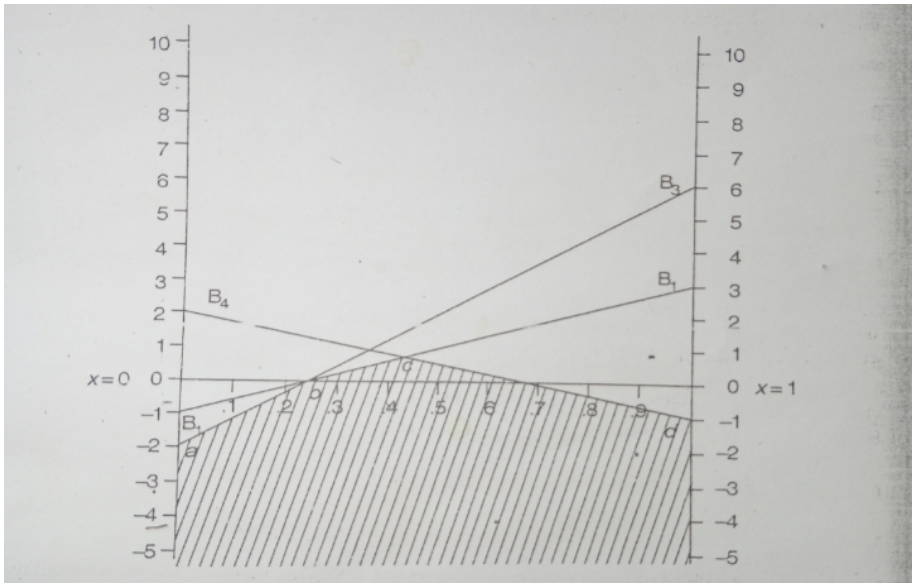

Figure 3.1: Graph with A's payoff function

Since, A is a maximin type player, identify the highest intersection point in the lower boundary of the graph. The lower boundary consists of the intersection points a,b,c and d. Out of these points, point c is at the highest level. Hence, the corresponding solution is the optimal solution to the given problem.

The line B_1 and B_4 pass through the point C. Hence, form a 2×2 payoff matrix for player A by retaining columns B_1 and B_4 is shown, with oddments of the rows and columns also also shown below.

	Player B		
Player A	1	4	oddments
1	3	-1	3
2	-1	-2	4
oddments	3	4	

Table 3.19: 2×2 payoff matrix with oddments

Let p_i be the probability of selection of alternative i by player A where $i = 1, 2$ and q_j be the probability of selection of alternative j by player B, where $j = 1, 2$. Then,

$$p_1 = \frac{3}{3+4} = \frac{3}{7}$$

$$p_2 = \frac{4}{3+4} = \frac{4}{7}$$

$$q_1 = \frac{3}{3+4} = \frac{3}{7}$$

$$q_2 = \frac{4}{3+4} = \frac{4}{7}$$
$$V = \frac{3(3) - 1(4)}{3+4} = \frac{5}{7}$$

The strategies of player A ans B, and the value of the game are:
A($\frac{3}{7}$, $\frac{4}{7}$) ; B($\frac{3}{7}$, 0 , 0, $\frac{4}{7}$) and $V = \frac{5}{7}$

Example 3.6.2.

Consider the playoff matrix of player A and solve it optimally using the graphical method.

	Player B				
Player A	1	2	3	4	5
1	3	6	8	4	4
2	-7	4	2	10	2

Solution: Here the maximin value is equal to the minimax value, equal to 3. Hence the game as a saddle point, as a result each player has a pure strategy and the corresponding results as shown below.
$A(1, 0)$, $B(1, 0, 0, 0, 0)$, $V = 3$.
Column 2 is dominated by column 5, hence column 2 is deleted.

	Player B			
Player A	1	3	4	5
1	3	8	4	4
2	-7	2	10	2

Table 3.20: Payoff matrix after deleting column 2

Let, x be the probability of selection of alternative 1 by player A and $1 - x$ be the probability of selection of alternative 2 by player A. Therefore the expected payoff to player A with respect to different alternatives of player B is summarized below 3.6.

B's alternative	A's payoff function
1	$3x - 7(1 - x) = 10x - 7$
3	$8x + 2(1 - x) = 6x + 2$
4	$4x + 10(1 - x) = -6x + 10$
5	$4x + 2(1 - x) = 2x + 2$

Table 3.21: Expected functions of player A

The computations of the expected payoff of player A to each of the alternatives of player B, when $x = 0$ as well as when $x = 1$ are summarized below.

B' alternative	A's expected payoff function	A's expected gain $x = 0$	$x = 1$
1	$10x - 7$	-7	3
3	$6x + 2$	2	8
4	$-6x + 10$	10	4
5	$2x + 2$	2	4

Table 3.22: Expected gain of player A

The expected gain function of player A with respect to different alternatives of player B are plotted in the graph below.

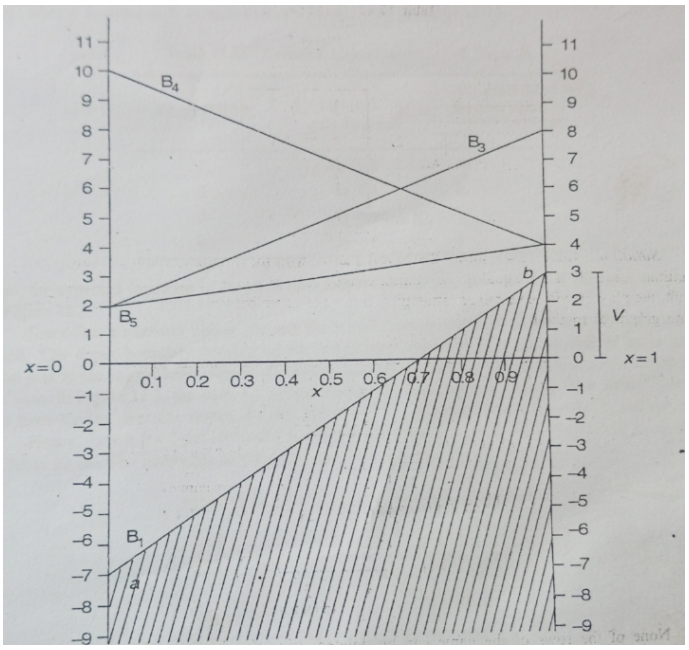

Figure 3.2: Graph with A's payoff

Since, A is a maximin player, identify the highest intersection point in the lower boundary of the graph. The lower boundary consists of the intersection points a and b. Out of these two points, b is at the highest level.That is, the corresponding solution is the optimal solution to the given problem. With respect to the intersection point b, value of the game is 3(the vertical distance of the intersection$b$ from the horizontal line).

The probability of selection of alternative 1 by player A is 1(i.e. $x = 1$). This is the horizontal distance to the point b from the vertical line. Therefore, the probability of selection of alternative 2 by player A is 0(i.e. $1 - x = 0$).

Since the maximin intersection point b is at the rightmost vertical line, the line

B_1 is the only line of player B passing through it. So, the probability of selection of alternative 1 by player B is 1 and the probabilities of selecting other remaining alternatives by player B are zero.

3.7 Linear Programming (LP) Approach for Game Theory

The games can be classified into the following types:

(i) Generalized game of size m×n with a saddle point

(ii) 2×2 game with or without saddle point

(iii) 2×n game without a saddle point

(iv) m×2 without a saddle point

(v) 3×3 game and games of higher sizes without saddle point

The first four(4) games can be solved using the methods discussed in earlier sections. But, the games under the 3×3 game and games of higher sizes without saddle point need a specializes procedure, namely *linear programming (LP) method* [11].

3.7.1 Development of LP model with respect to player A

The expected gain to A with respect to the selection of each of the alternatives of B is presented below.

$B'sAlternative$	Expected gain function to Payer A
1	$a_{11}p_1 + a_{12}p_2 + \cdots \quad +a_{i1}p_i + \quad \cdots \quad +a_{1m}p_m = \sum_{i=1}^{m} a_{i1}p_i$
2	$a_{21}p_1 + a_{22}p_2 + \cdots \quad +a_{i2}p_i + \quad \cdots \quad +a_{m2}p_m = \sum_{i=1}^{m} a_{i2}p_i$
$\vdots$	$\vdots \quad \vdots \quad \vdots \quad \vdots$
j	$a_{1j}p_1 + a_{2j}p_2 + \cdots \quad +a_{ij}p_i + \quad \ldots \quad +a_{mj}p_m = \sum_{i=1}^{m} a_{ij}p_i$
$\vdots$	$\vdots \quad \vdots \quad \vdots \quad \vdots$
n	$a_{1n}p_1 + a_{2n}p_2 + \cdots \quad +a_{in}p_i + \quad \cdots \quad a_{mn}p_m = \sum_{i=1}^{m} a_{in}p_i$

(3.2)

Descriptive model for player A: Since, A is a maximin player, he tries to maximize the value of the game. This is achieved by maximizing the minimum of the functions given in the last column of the table above .

$$\text{Max}\Big[Min\Big(\sum_{i=1}^{m} a_{i1}p_i, \sum_{i=1}^{m} a_{i2}p_i, ..., \sum_{i=1}^{m} a_{ij}p_i, ..., \sum_{i=1}^{m} a_{in}p_i\Big)\Big]$$

Subject to;

$$p_1 + P_2 + ... + p_i + ... + p_m = 1$$

Also, $p_1 + P_2 + ... + p_i + ... + p_m \geq 0$

Since the objective function of this model is not linear, it is called a descriptive model. The above model is converted into a linear model since it is impossible to work with descriptive models.
Let,
V=$Min\Big(\sum_{i=1}^{m} a_{i1}p_i, \sum_{i=1}^{m} a_{i2}p_i, ..., \sum_{i=1}^{m} a_{ij}p_i, ..., \sum_{i=1}^{m} a_{in}p_i\Big)$
The linear model with respect to the above substitution is:

Maximize Z=V

Subject to,

$$\sum_{i=1}^{m} a_{i1}p_i \geq V$$

$$\sum_{i=1}^{m} a_{i2}p_i \geq V$$

$$\vdots$$

$$\sum_{i=1}^{m} a_{ij}p_i \geq V$$

$$\vdots$$

$$\sum_{i=1}^{m} a_{in}p_i \geq V$$

$$p_1 + p_2 + ... + p_i + ... + p_m = 1$$

$p_i \geq 0$, $i = 1, 2, ..., m$
The system constraints of the above model can be simplified by dividing it by V as follows:

$$\sum_{i=1}^{m} \frac{a_{i1}p_i}{V} \geq 1$$

$$\sum_{i=1}^{m} \frac{a_{i2}p_i}{V} \geq 1$$

$$\sum_{i=1}^{m} \frac{a_{ij}p_i}{V} \geq 1$$

$$\sum_{i=1}^{m} \frac{a_{in}p_i}{V} \geq 1$$

$\frac{P_1}{V}+\frac{P_2}{V}+.....+\frac{P_i}{V}+...+\frac{P_m}{V} = \frac{1}{V}$

In this model;
(i) If the value of the game is less than zero, the type of each constant will get changed;
(ii) If the value of the game is zero, the terms of the constraints will become infinite

Therefore, to avoid these problems, each of the entries of payoff matrix is added with a constant K which is equal to the absolute value of the maximum of the negative values of the payoff matrix plus one.
The true value of the game is obtained by subtracting K from the values of the game. If all the values of the payoff matrix are greater than zero, there is no need to add a constant to each of the entries.

Let, $\frac{P_i}{V} = X_i$, $i = 1, 2, ..., m$. Therefore,

$maxV = min\frac{1}{V}$
$=\max\left(\frac{P_1}{V} + \frac{P_2}{V} + ... + \frac{P_i}{V} + ... + \frac{P_m}{V} = \frac{1}{V}\right)$
$=\min\left(X_1 + X_2 + ... + X_i + ... + X_m\right)$

Then, the revised model is;

$\sum_{i=1}^{m} a_{i1}X_i \geq 1$

$\sum_{i=1}^{m} a_{i2}X_i \geq 1$
$\vdots$
$\sum_{i=1}^{m} a_{ij}X_i \geq 1$
$\vdots$
$\sum_{i=1}^{m} a_{in}X_i \geq 1$

$X_i \geq 0$, $i = 1, 2, ..., m$

The values of p_i are obtained using the following formulae.

$V = \frac{1}{Z_i}$

$p_i = VX_i,\ i = 1, 2, ...m$

3.7.2 Development of LP model with respect to player B

he expected loss(+)/gain(-) to player b with respect to the selection of each of the alternatives of player is presented below.

$A'sAlternative$	Expected loss(+)/gain(-) function to Player B			
1	$a_{11}q_1 + a_{12}q_2 + \cdots$	$+a_{j1}q_j+$	$\cdots$	$+a_{1n}q_n = \sum_{j=1}^{n} a_{j1}q_j$
2	$a_{21}q_1 + a_{22}q_2 + \cdots$	$+a_{j2}q_j+$	$\cdots$	$+a_{n2}q_n = \sum_{j=1}^{n} a_{j2}q_j$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
i	$a_{i1}q_1 + a_{i2}q_2 + \cdots$	$+a_{ij}q_j+$	$\ldots$	$+a_{in}q_n = \sum_{j=1}^{n} a_{ij}q_j$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
m	$a_{1m}q_1 + a_{2m}q_2 + \cdots$	$+a_{mj}q_j+$	$\cdots$	$a_{mn}q_n = \sum_{j=1}^{n} a_{jm}q_j$

(3.3)

The descriptive model for player B and its revised model can computed and presented as in for player A, keeping in mind that B is a minimax player,hence he/she tries to minimize the value of the game. This is done by minimizing the maximum of the functions given in the last column 3.7.1.

Example 3.7.3.

Consider the payoff matrix with respect to player A and solve it optimally using LP.

	Player B		
Player A	1	2	3
1	1	-1	-1
2	-1	-1	3
3	-1	2	-1

Solution: Here, the maximin value(-1) is not equal to the minimax value (1). The game has no saddle point, and in turn the game is said to have pure strategies. Also, one can verify the fact that the payoff matrix cannot be reduced using dominance property. Hence the game should be solved using linear programming (LP).
Since the payoff matrix has negative values, the absolute value of the most negative value plus 1, i.e. $(K = 1 + 1)$ is added to each of the entries of the payoff matrix the corresponding payoff matrix below.

	Player B		
Player A	1	2	3
1	3	1	1
2	1	1	5
3	1	4	1

Since the LP formulation with respect to player B will have only "$\leq$" constraints, it is advisable to develop a mode for player B. The solution of the problem can be obtained in the following ways:

(i) Solve the model and obtain the strategies of B and the value of the game.

(ii) Obtain the strategies of A from the optimal table of B using the concept of duality.

(iii) The true value of the game is obtained by subtracting K from the value of modified game.

The last step os required because $K(2)$ is added to all entries of the payoff matrix at the initial stage. Let a_{ij} be the payoff to player A. If A selects alternative i and B selects alternative j. V be the value of the game(i.e. expected gain for player A. So, A will try to maximize$V$ and B will try to minimize V), p_i be the probability of selection of alternative i by player A and q_j be the probability of selection of alternative j by player B, where $i = 1, 2, 3$ and $j = 1, 2, 3$.

Development of LP model with respect to player B. The expected loss(+)/gain(-) function to B with respect to the selection of each alternatives of A is presented below along the necessary condition of $q_1 + q_2 + q_3 = 1$

$$3q_1 + q_2 + q_3 \leq V$$

$$q_1 + q_2 + 5q_3 \leq V$$

$$q1 + 4q_2 + q_3 \leq V$$

$$q_1 + q_2 + q_3 = 1$$

Dividing the set of constraints by V, we get,

$$\frac{q_1}{V} + \frac{q_2}{V} + 5\frac{q_3}{V} \leq 1$$

$$\frac{q_1}{V} + 4\frac{q_2}{V} + \frac{q_3}{V} \leq 1$$

$$\frac{q_1}{V} + \frac{q_2}{V} + \frac{q_3}{V} = \frac{1}{V}$$

Substituting q_j for Y_j in the above systems of constraints , we get

$$3Y_1 + Y_2 + Y_3 \leq 1$$

$$Y_1 + Y_2 + 5Y_3 \leq 1$$

$$Y_1 + 4Y_2 + Y_3 \leq 1$$

$$Y_1 + Y_2 + Y_3 = \frac{1}{V}$$

As per the guidelines for constructing LP model for player B, the LP model from the above systems of constraints is presented below.

Maximize $Z_2 = Y_1 + Y_2 + Y_3$

Subject to

$$\begin{aligned} 3Y_1 + Y_2 + Y_3 &\leq 1 \\ Y_1 + Y_2 + 5Y_3 &\leq 1 \\ Y_1 + 4Y_2 + Y_3 &\leq 1 \\ Y_1, Y_2, Y_3 &\geq 0 \end{aligned}$$

Converting the above generalized model into a standard model yields:

Maximize $Z_2 = Y_1 + Y_2 + Y_3 + 0S_1 + 0S_2 + 0S_3$
Subject to;

$$\begin{aligned} 3Y_1 + Y_2 + Y_3 + S_1 &= 1 \\ Y_1 + Y_2 + 5Y_3 + S_2 &= 1 \\ Y_1 + 4Y_2 + Y_3 + S_3 &= 1 \\ Y_1, Y_2, Y_3, S_1, S_2, S_3 &\geq 0 \end{aligned}$$

Example 3.7.4.

Suppose that you are a politician trying to win an election. Town Q has three(3) different types of areas; Urban, sub-urban and Rural. These areas have respectively 100000 , 200000 and 50000 registered voters.
To govern effectively, you would like to win the majority of the votes in each of the 3 regions. You are honorable and would never consider supporting policies in

which you do not support. You realize, however the certain issues may be more effective in winning votes in certain places. Your primary issues are building more roads, gun control, farm subsidies and petroleum(gasoline) tax dedicated to improve public transport.[11]
According to your campaign staff's research , you can can estimate hoe many votes you win or loose from each population segment by spending $1,000 on advertising on each issue. The information appears in the table below [13] .

	Areas		
Policy	Urban	Sub-urban	Rural
Build roads	-2	5	3
Gun control	8	2	-5
Farm subsidies	0	0	10
Gasoline Tax	10	0	-2

Table 3.23:

In this table, each entry describes the number of thousands of either urban, sub-urban or rural voters which could be won over by spending $1000 on advertising in support of a particular issue. Negative entries denote votes that would be lost. Your task is to figure out the minimum amount of money you need to spend in order to win 50000 urban votes, 100000 sub-urban votes and 25000 rural votes.
By trial and error, it is possible to come up with a strategy that will win the required number of votes, but such a strategy may not be the least expensive one.
Let X_1, X_2, X_3, X_4 be the number of thousands of dollars spent on advertising on building roads, gun control, farm subsidies and gasoline(petrol) tax.
We can write the requirement that we win at least 50000 urban votes as;

$-2X_1 + 8X_2 + 0X_3 + 10X_4 \geq 50$
Similarly, we can write the requirement that we win at least 100000 sub-urban votes and 25000 rural votes as;

$5X_1 + 2X_2 + 0X_3 + 0X_4 \geq 100$

$3X_1 - 5X_2 + 10X_3 - 2X_4 \geq 25$
We would like to minimize the expression,
$X_1 + X_2 + X_3 + X_4$
$X_1 \geq 0$, $X_2 \geq 0$, $X_3 \geq 0$, $X_4 \geq 0$

Solution
Minimize Z=$X_1 + X_2 + X_3 + X_4$
Subject to;

$-2X_1 + 8X_2 + 0X_3 + 10X_4 \geq 50$

$5X_1 + 2X_2 + 0X_3 + 0X_4 \geq 100$

$3X_1 - 5X_2 + 10X_3 - 2X_4 \geq 25$

X_1 , X_2, X_3, $X_4 \geq 0$ each.

The next chapter explains the analysis of data; the analysis of two political parties in Zambia using game theory and LP.

4 Data Analysis

Conflicts and competitions are part of life. Examples are Military battles, advertisement by similar business entities, campaigns by political parties etc. According to Boah et[3], scientific fashions of this nature formally referred to as game theory can be handled in a well-organized way. Competition in politics is seen when two or more candidates or parties strive for a goal which cannot be shared through an election [2]. Also the winner gets the seat of the elected office for a predefined period of time after which another election is conducted to determine the next holder of the office. Cadman [4] presented that manifestos normally provide an opportunity to make proposed policies or promises and interventions known to electorates so as to win their votes. Again, manifestos can provide tangible means for voters to assess the capabilities of parties which aim to solicit as many voters as possible. Game theory, a multidisciplinary framework, offers a profound understanding of strategic decision-making, illuminating the intricate dynamics of competition ans cooperation. It gives a robust toolkit for for analyzing complex interactions among rational actors, enabling the prediction of optimal outcomes.

Democracy is consolidating across across Africa. An outstanding feature of democracy is that, the power to rule lies with the people. Democracy in Zambia is very vibrant and elections are keenly contested. Generally, the elections are credible, and almost all voters participate in them. The winning party is in charge of everything. Only four political parties in Zambia, the United National Independence Party (UNIP), Movement for Multi-party Democracy (MMD), the Patriotic Front (PF) and the United Party for National development (UNPD) have had the opportunity to be in power since 1964 in spite of the many political parties in the country. The two dominant or major political parties are characterized by their ideologies. The UPND claims to be socially democratic whilst the PF, liberally democratic.

Due to the competitive nature of politics in Zambia, political parties in the country always have issues to deal with. They are always concerned with how to keep or maintain their existing support base and also win frustrated members of the opposing parties or win floating voters to their parties. Also every political party in the country is challenged with rolling policies and strategies that will please electorates so as to lure them to like their parties.
Game theory which has an impact on almost every competition can be employed to obtain optimal ways of operation for political parties in the country. When the best or optimal ways are identified, the trust of political parties will improve and citizens' satisfaction will be increased.
The general objective of the study was to analyze the performance of two dominant or major political parties since 2011 using Game Theory and Linear Programming. Specifically, the study sought to construct a payoff matrix of the performance of two political parties using game theory; formulate the payoff matrix as a linear programming (LP) problem and solve; and finally determine optimal strategies for for the two political parties.

4.1 Materials and Methods

The study mainly used primary data which was collected using a questionnaire prepared and administered to two hundred (200) educated respondents to find their patronage for the two political parties as far as eight (8) selected sectors of governance are concerned. The sectors of governance were regarded or taken as strategies used by the two political parties to influence citizens to patronize their parties in Zambia. Due to the swing nature (per the electoral history) of the Lusaka and The Copperbelt provinces of Zambia respectively, they were chosen purposively for the study. The respondents were sampled purposively by considering their ages and how politically unbiased their environment have been. Specifically, the technique was employed to obtain varied views of citizens in swing regions in the country who have witnessed elections from 2006 to date. Respondents who needed further explanation of the questionnaire were assisted in that direction.
In game theory, an opponent is referred to as a player 3.2. Each player has a number of choices, finite of infinite called strategies. the outcomes or payoffs of a game are summarized as functions of different strategies for each player where if one player wins, another player looses, the game ia called zero-sum game 3.2. Some of the procedures/methods usually used in game theory are Maximin-Minimax method, Dominance method 3.5, Graphical method and Linear Programming 3.6.

4.2 Results and discussions

The eight (8) purposively selected sectors of governance for the study are given in the table.
The row player (UPND) and the column player (PF) are the players of two-person constant-sum game with the constant being 200 respondents. As already indicated, in a two-person constant-sum game, a matrix is used to summarize the payoffs to the row player and the sum of payoffs to both players is always constant. From the responses of the two hundred (200) selected people for the study, a corresponding payoff matrix for the row player (UPND) has been constructed as given below.

Table 4.1: Sectors of Governance selected for the study

Sector	Abbreviation	Sector	Abbreviation
Health	H	Education	ED
Energy	EG	Agriculture	AG
Employment and Labor	EL	Youth and Sports	YS
Petroleum	P	Justice	J

	ED	EL	YS	J	P	H	AG	EG	Row. Min.
ED	**124**	86	92	95	98	87	83	92	**83**
EL	89	**138**	82	98	89	91	86	91	**82**
YS	92	86	**89**	83	85	94	80	97	**80**
J	94	87	83	**97**	92	86	97	86	**83**
P	82	99	89	91	**94**	94	88	93	**82**
H	90	83	84	84	93	**94**	80	90	**80**
AG	88	98	94	81	82	91	**102**	89	**81**
EG	94	93	95	85	98	83	92	**96**	**83**
Col. Max.	**124**	**138**	**95**	**97**	**98**	**94**	**102**	**98**	

The table above shows the payoff matrix for the row player UPND. From the table the following deductions where made:

(i) 124 preferred Education (ED) of the row player (UPND) and 200-124= 76 preferred Education (ED) for the column player (PF). Similarly, 138, 89, 97, 94, 94, 102, 96 opted for Employment and labor (EL), Youth and Sports (YS) , Justice (J), Petroleum (P), Health (H), Agriculture (AG), and Energy (EG), respectively from the row player (UPND) whereas, 62, 111, 90, 106, 106, 98 and 104 respondents respectively opted for the same sectors from the column player (PF).

(ii) The elements in the main diagonal or the principal namely 124, 138, 89, 97, 94, 94, 102 and 96 indicate that, the respondents see the row player (UPND) to have performed better in Education (ED), and Employment, Justice (J), Agriculture (AG) and Employment and Labor (EL) than the column player (PF) whereas the column player (PF) are seen to have performed better in the remaining sectors of governance than row player (UPND).

(iii) The row player (UPND) was preferred to the column player (PF) in comparing their performance in Employment against Education and Employment and labor (EL), Agriculture (AG), and also Employment and Labor against Education (ED). On the other hand, the column player (PF) was preferred to the row player UPND in the rest of the comparisons.

(iv) The maximum of the row values (Maximin)=83 and the minimum of the column values (Minimax)= 94. Since the Maximin $\neq$ Minimax. The game has no saddle point. Therefore, the value of the game, V to the row player UPND cannot be readily obtained. Linear programming method which is capable of handling mixed-strategy games of larger dimensions is therefore employed to formulate the payoff matrix(table above) as a Linear Programming model to obtain the value of the game V to the row player UPND.

Resulting Linear Programming model for the Row player UPND

Maximize $\frac{1}{V} = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8$

Subject to

$124x_1 + 89x_2 + 92x_3 + 94x_4 + 82x_5 + 90x_6 + 88x_7 + 94x_8 \geq 1$

$86x_1 + 138x_2 + 86x_3 + 87x_4 + 99x_5 + 83x_6 + 98x_7 + 93x_8 \geq 1$

$92x_1 + 82x_2 + 89x_3 + 83x_4 + 89x_5 + 84x_6 + 94x_7 + 95x_8 \geq 1$

$95x_1 + 98x_2 + 83x_3 + 100x_4 + 91x_5 + 84x_6 + 81x_7 + 85x_8 \geq 1$

$98x_1 + 89x_2 + 85x_3 + 92x_4 + 94x_5 + 93x_6 + 82x_7 + 98x_8 \geq 1$

$87x_1 + 91x_2 + 94x_3 + 86x_4 + 94x_5 + 94x_6 + 91x_7 + 83x_8 \geq 1$

$83x_1 + 86x_2 + 80x_3 + 97x_4 + 88x_5 + 80x_6 + 102x_7 + 92x_8 \geq 1$

$92x_1 + 91x_2 + 97x_3 + 86x_4 + 93x_5 + 90x_6 + 89x_7 + 98x_8 \geq 1$

$x_i = \frac{P_i}{V} \geq 0,\ i = 1, 2, ..., 8$

Where p_i is the probabilities of the row player UPND selecting the 8 strategies(sectors) and V is the value of the game for the row player.

5 Conclusion

This Thesis has demonstrated the efficacy of game theory in analyzing and predicting political phenomena. By applying Game-theoretic concepts, coalition formation and international negotiations, this research has provided valuable insights into the strategic interactions that shape political outcome.
The findings of this study underscore the importance of considering the rational, self-interested behavior of political actors when analyzing complex decision-making process. This study has contributed significantly to knowledge by providing scientific ways by which political parties should consider game theory when analyze their strategies to become vibrant and competitive. Also, the study has filled the knowledge or research gap of application of game theory to politics in the manner presented in this paper which appeared non-existent.

The performance of two political parties since 2011 have been successfully investigated and using Game Theory and Linear Programming. Specifically, a payoff matrix of the performance of the two major political parties since 2011 has been successfully constructed using Game theory; the values of the game theory associated with the payoff matrix have been successfully obtained using the concept of linear programming and finally, for the two political to become more vibrant and competitive, their optimal strategies have been clearly spelt out.

5.1 Recommendations

I recommend that the two political parties should maintain their perceived good performance in the indicated sectors and improve upon their perceived bad performance in the remaining sectors of governance. Also , all political parties should consider game theory when analyzing their political strategies. Future research should consider exploring game theory's applications in emerging technologies(e.g. block chain), investigating its role in addressing global challenges such as climate change and weather. Also to foster a culture of strategic thinking , encouraging policy markers to consider the long-term consequences of their actions.

5.2 Limitations

(i) Oversimplification: Game theory models oversimplify complex political phenomena.

(ii) Assumption of Rationality: Game theory assumes rational actors, but real world decision-making is often influenced by emotions, biases, and cognitive limitations.

(iii) Data Quality: Game theory models rely on high quality of data, which may be difficult to obtain in political contexts.

(iv) Data volume: Large datasets require significant time and resources to collect and process.

(v) Could not solve and find the value of V because of time and financial constraints

6 Appendix

The following questionnaire was used to collect data from the people. The instructions/procedures followed by respondents are as follows;

(i) Kindly indicate your age range ticking one of the given ranges of ages.

(ii) Tick a cell after comparing the respective sectors of governance that make up that cell if and only if the UNPD in Zambia have performed better than the PF.

(iii) Leave a cell blank after comparing the respective sectors of governance that make up that cell if and only if the PF in Zambia have performed better that the UPND in Zambia.

Section A

(i) Which political party have performed better in Employment and Labor?

○ UPND
○ PF

(ii) Which political party have performed better in Energy?

○ UPND
○ PF

(iii) Which political party have performed better in Education?

○ UPND
○ PF

(iv) Which political party have performed better in Health?

○ UPND
○ PF

(v) Which political party have performed better in Agriculture?

○ UPND
○ PF

(vi) Which political party have performed better in Justice?

○ UPND
○ PF

(vii) Which political party have performed better in Petroleum?

○ UPND
○ PF

(viii) Which political party have performed better in Youth and Sports?

- ○ UPND
- ○ PF

Section B: Demographic information

Kindly indicate your Age range by ticking on the given ranges of ages.

(i) ○ under 20

(ii) ○ 20 to 30

(iii) ○ 31 to 40

(iv) ○ 41 to 50

(v) ○ over 50

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